lymphoCYB

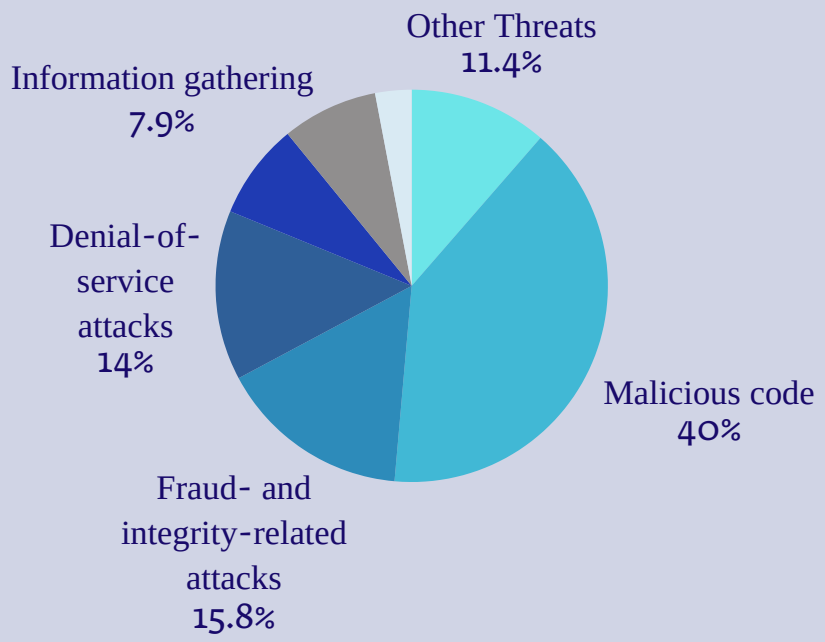
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Introduction

Tools and services in the industrial cybersecurity sector have become numerous and diverse, and it is rare to find a single tool that covers all security requirements—particularly those that verify system compliance with the standards and controls specific to the Kingdom of Saudi Arabia.

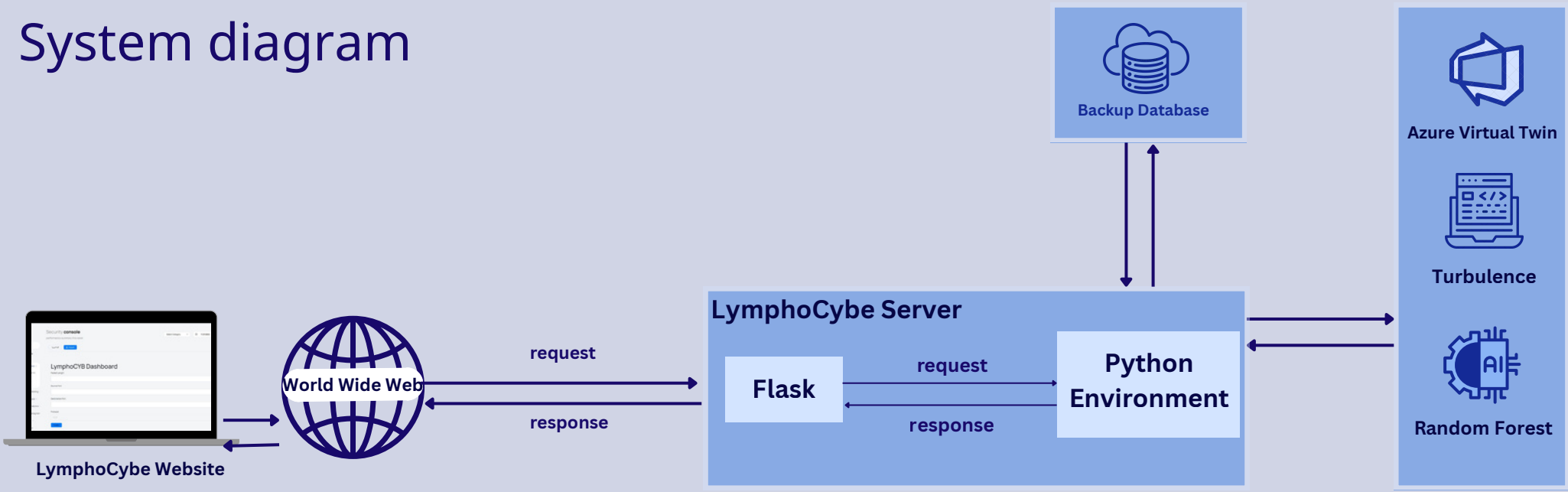
"LymphoCybe" draws inspiration from the immune system in divine creation, where white blood cells learn how to defend against a virus when encountering it again. This concept is realized by leveraging artificial intelligence and the ATT&CK knowledge base.

Statistics



The chart illustrates the threats faced by a SCADA system.

System diagram



The client selects the desired operation from the following options: system assessment, attack simulation, intentionally failing training scenarios, or verification of the system's compliance with the standards of the National Cybersecurity Authority. This is done by sending a request to a "Flask" server, which is then managed through a "Python" environment. The "Azure Virtual Twin" simulates the factory system by leveraging backup copies. "Turbulence" runs scenarios with intentional operational errors to identify sensitive components within the system. "Random Forest" analyzes network packets and identifies affected ones, achieving an accuracy rate of up to 100%.

Future Work

1. Building specialized artificial intelligence models
2. Integration with factory protocols such as MODBUS
3. Adding automation features, including developing an automated response system that isolates devices affected by attacks and scheduling periodic reports to be sent to network administrators
4. Building specialized intentionally failing scenarios

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Reffrences